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Evaluating Morphological Alignment of Tokenizers in 70 Languages

Section header 1

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Section header 2

Abstract

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Tokenization is a key step in language modeling, with effects on model training and performance, it remains unclear how to effectively evaluate tokenizer quality. One proposed dimension of tokenizer quality is the extent to which tokenizers preserve linguistically meaningful subwords, aligning token boundaries with morphological boundaries within a word. We expand MorphScore (Arnett & Bergen, 2025), which previously covered 22 languages, to support a total of 70 languages. The updated MorphScore offers more flexibility in evaluation and addresses some of the limitations of the original version. We then correlate our alignment scores with downstream task performance for five pre-trained language models on seven tasks, with at least one task in each of the languages in our sample. We find that morphological alignment does not explain very much variance in model performance, suggesting that morphological alignment alone does not measure dimensions of tokenization quality relevant to model performance.

 MorphScore evaluator

 datasets  code and data

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as it would enable tokenizer selection before model training, leading to significant computational and financial savings.

One of the most frequently used intrinsic tokenizer evaluations is compression. Compression is often measured as the number of tokens it takes to encode a text given a particular tokenizer. It is relatively easy to measure, as it requires simply tokenizing a text and calculating token counts. One metric of compression is fertility, i.e. the number of tokens per word (Rust et al., 2021). Fertility is simple to implement but can be difficult to generalize crosslinguistically, as wordhood is often operationalized as whitespace-separated orthographic units. Not all languages use whitespaces, e.g. Mandarin Chinese, Thai, and Khmer. Corpus token count (CTC; Schmidt et al., 2024) is the total tokens it takes to represent a text for a given tokenizer. CTC can be compared, therefore across tokenizers of different types, vocabulary sizes, etc. It has also been used to compare compression crosslinguistically, by calculating CTC over parallel text in order to determine crosslinguistic differences in compression (Arnett & Bergen, 2025).

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Text 9 have argued that increased compression increases the token density for a sequence of fixed length, which leads to improved model performance (Deletang et al., 2024). There has been empirical evidence to support the claim that more tokenizer compression is correlated with better task performance (Goldman et al., 2024; Gallé, 2019). However, more recent work has shown that there is no robust relationship between tokenizer compression and language model performance (Schmidt et al., 2024).

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Text 17 intrinsic tokenizer evaluations have been proposed. Text 18 Rényi efficiency (Zouhar et al., 2023), which Text 19 account frequency distribution. More optimal Rényi efficiency is associated with having more compression for higher-frequency items and less compression for lower-frequency items. Zouhar et al. (2023) released the Text 20 tokenization-scorer package to support calculation of Text 21 efficiency for any tokenized text. However, later Text 22 argues it may not provide a holistic metric of good Text 23 tokenization quality (Cognetta et al., 2024).

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Text 24 property of tokenizers that has been studied is how Text 25 morphologically aligned tokenization is, or to what extent Text 26 boundaries align with morpheme boundaries for a given word. For example, the English word ‘books’ is

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1. Introduction

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Tokenization is the first step of language modeling, in which strings of text are segmented into discrete units in the tokenizer’s vocabulary. Tokenization has been shown to have effects on speed and efficiency of language model training (Dagan et al., 2024; Ali et al., 2024; Asgari et al., 2025), performance (Ali et al., 2024), and inference cost and latency (Ahia et al., 2023; Petrov et al., 2023). Despite this, it is still unclear how to best evaluate tokenizers. Finding reliable intrinsic tokenizer evaluation would be enormously valuable.

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